

Question 1 (2 points)

The class NP is defined as the class of problems that

- A. Can be solved in polynomial time by a nondeterministic Turing Machine
- B. Cannot be solved in polynomial time by a deterministic Turing Machine
- C. Can be solved in polynomial time by a deterministic Turing Machine
- D. Cannot be solved in polynomial time by a nondeterministic Turing Machine

Option A is correct. Refer to section 18.3.2 of Week 8 Lecture Notes.

Question 2 (2 points)

The class coNP is defined as

- A. $\{L \mid \bar{L} \in \text{NP}\}$
- B. $\{L \mid L \notin \text{NP}\}$
- C. $\{\bar{L} \mid L \subseteq L' \in \text{NP}\}$
- D. $\{\bar{L} \mid L \notin \text{NP}\}$

Option A is correct. Refer to section 18.3.4 of Week 8 Lecture Notes.

Question 3 (2 points)

$L \in \text{NP}$ if there exists constants $c > 0$ and $k \geq 0$, and a polynomial time Turing Machine V , such that

- A. $\exists x \in \Sigma^*, x \in L \iff \forall y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$
- B. $\forall x \in \Sigma^*, x \in L \iff \exists y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$
- C. $\forall x \in \Sigma^*, x \in L \iff \forall y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$
- D. $\exists x \in \Sigma^*, x \in L \iff \exists y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$

Option B is correct. Refer to section 18.3.2 of Week 8 Lecture Notes.

Question 4 (2 points)

$L \in \text{coNP}$ if there exists constants $c > 0$ and $k \geq 0$, and a polynomial time Turing Machine V , such that

- A. $\exists x \in \Sigma^*, x \in L \iff \forall y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$
- B. $\forall x \in \Sigma^*, x \in L \iff \exists y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$
- C. $\forall x \in \Sigma^*, x \in L \iff \forall y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$
- D. $\exists x \in \Sigma^*, x \in L \iff \exists y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$

Option C is correct. Consider the definition 18.3.1 given in section 18.3.2 of Week 8 Lecture Notes.

$L \in \text{NP}$ if there exists constants $c > 0$ and $k \geq 0$, and a polynomial time Turing Machine V , such that

$$\forall x \in \Sigma^*, x \in L \iff \exists y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$$

The above can be modified as follows

$L \in \text{coNP}$, that is, $\bar{L} \in \text{NP}$ if there exists constants $c > 0$ and $k \geq 0$, and a polynomial time Turing Machine V' , such that $\forall x \in \Sigma^*$,

$$x \in \bar{L} \iff \exists y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V'(x, y) = 1$$

$$\text{i.e. } x \notin L \iff \exists y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V'(x, y) = 1$$

$$\text{i.e. } \neg(x \notin L) \iff \neg(\exists y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V'(x, y) = 1)$$

$$\text{i.e. } (x \in L) \iff \forall y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V'(x, y) \neq 1$$

Now, if we consider V to be the Turing Machine that accepts the complement of the language accepted by V' , then we get the following statement.

$L \in \text{coNP}$ if there exists constants $c > 0$ and $k \geq 0$, and a polynomial time Turing Machine V , such that

$$\forall x \in \Sigma^*, (x \in L) \iff \forall y \in \Sigma^* \text{ such that } |y| \leq c|x|^k \text{ and } V(x, y) = 1$$

Question 5 (2 points)

Which of the following problems is not known to be in the complexity class P?

- A. Deciding whether a natural number is even
- B. Deciding whether a boolean formula is satisfiable**
- C. Deciding whether a graph is connected
- D. Deciding whether a binary string has even number of 1's

Option B is correct. This problem, also known as SAT, is an NP-complete problem. It is not known whether NP-complete problems can be solved deterministically in polynomial time or not.

Question 6 (2 points)

Which of the following problems is known to be in the complexity class coNP?

- A. Deciding whether a natural number is prime**
- B. Deciding whether a boolean formula is satisfiable
- C. Deciding whether a graph has a Hamiltonian path
- D. Deciding whether two graphs are isomorphic

Option A is correct. The complement of this problem, that is, deciding whether a natural number is composite is in NP. Refer to section 18.3.3 of Week 8 Lecture Notes.

Question 7 (2 points)

Let L be a language in the complexity class P. Which of the following is not known?

- A. $L \in \text{NP}$
- B. $L \in \text{coNP}$
- C. $L \in \text{NP} \cap \text{coNP}$
- D. $L \in \text{NP-complete}$**

Option D is correct. It is not known whether any problem in P (that is a problem that can be solved deterministically in polynomial time) is also in NP-complete. In fact, if that happens, then it will be proved that $\text{P} = \text{NP}$, which is a very famous open question.

Question 8 (2 points)

If $L \in \text{NTIME}(f(n))$, then

- A. $L \in \text{DTIME}(f(n))$.
- B. $L \in \text{DTIME}(\log f(n))$.
- C. $L \in \text{DTIME}(f(n)^2)$.
- D. $L \in \text{DTIME}(2^{O(f(n))})$.**

Option D is correct. Refer to section 20.1 of Week 8 Lecture Notes.

Question 9 (2 points)

The class NP is not known to be closed under which of the following operations?

- A. Union
- B. Intersection
- C. Concatenation
- D. Complementation**

Option D is correct. The complement of a language in NP belongs to the class coNP. If NP is closed under complementation, this would indicate $\text{NP} = \text{coNP}$, which is also a very famous open question.

Question 10 (2 points)

If $\text{coNP} \subseteq \text{NP}$ then which of the following is true?

- A. $\text{NP} = \text{coNP}$
- B. $\text{P} = \text{NP}$
- C. $\text{P} \neq \text{NP}$
- D. $\text{NP} \neq \text{coNP}$

Option A is correct.

Let us assume $\text{coNP} \subseteq \text{NP}$. Now, let $L \in \text{NP}$. Therefore,

$\bar{L} \in \text{coNP}$

$\implies \bar{L} \in \text{NP}$ (By our initial assumption)

$\implies L \in \text{coNP}$ (By definition of coNP)

This proves $\text{NP} \subseteq \text{coNP}$. Therefore, together with our initial assumption, this means $\text{NP} = \text{coNP}$.

Question 11 (2 points)

The class NP-hard is defined as the set of

- A. Problems which are reducible in polynomial time to some NP complete problem
- B. Problems which are reducible in polynomial space to some NP complete problem
- C. **Problems to which all problems in NP are reducible in polynomial time**
- D. Problems to which all problems in NP are reducible in polynomial space

Option C is correct. Refer to section 19.1 of Week 8 Lecture Notes.

Question 12 (2 points)

If a decision problem A is polynomial time reducible to another decision problem B , then which of the following statements is necessarily true?

- A. **If A is NP-hard then B is also NP-hard**
- B. If B is NP-hard then A is also NP-hard
- C. If A is in NP then B is in NP
- D. If A is NP-complete then B is also NP-complete

Option A is correct. Refer to section 19.2 of Week 8 Lecture Notes.